Assignment 2:
Data Analysis Report

36103: Statistical Thinking for Data Science

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Group Number: 5

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Executive Summary

Electricity demand in NSW changes with weather, time of day, and season. During peak periods, wholesale prices can reach $17,500/MWh, so reliable short-term forecasts matter for both cost management and grid stability.

This project used high-frequency NSW data from 2022-2026 to study demand patterns and build forecasting models. Exploratory analysis showed demand is lowest in mild conditions, rises during extreme hot or cold weather, and differs between weekdays and weekends. Rooftop solar reduces midday demand but causes a sharp evening increase.

CatBoost is the recommended model. Its average error of around 389 MW means forecasts should inform decisions, not drive them outright, particularly during extreme weather or supply disruptions.

CatBoost achieved the best test performance: MAPE 4.45%, Adjusted R^2 0.884, MAE 389 MW.

XGBoost followed closely at 5.16% and 0.860. Ridge Regression (10.3%, 0.375) was insufficient for grid operations.

Both tree models showed no meaningful performance drop from validation to test.

Temperature and hour of day ranked as top predictors in both ensemble models, matching the U-shaped demand curve and intraday peaks from EDA.

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1 Introduction

1.1 Problem Statement

Electricity supply and demand must balance every 30 minutes. When they don't, grid operators absorb the cost. NSW wholesale prices can reach $17,500/MWh under the market cap that applied through mid-2025 (AER, 2025b). Accurate short-term demand forecasting keeps dispatch decisions ahead of those spikes.

1.2 Rationale

Serving 11 million customers, the NEM relies on NSW as its primary generator (AER, 2026). However, the lack of storage makes the grid vulnerable to demand swings. NSW's 7.5 GW of rooftop solar (the highest globally per capita) creates a "duck curve" effect: suppressing midday demand before requiring a massive evening ramp from dispatchable generators (CEC, 2025; Green.com.au, 2026). This volatility drove 2024 wholesale prices up 43% to $150/MWh (AER, 2025a), making accurate forecasting critical to reducing costs.

1.3 Project Aims and Objectives

The aim of this project was to determine whether machine learning models trained on historical half-hourly NSW data can predict grid electricity demand accurately enough to support operational dispatch planning. To support this aim, we asked:

1. Does temperature have a non-linear relationship with NSW grid demand across 2°C bins from 5°C to 43°C?
2. Do mean hourly demand profiles differ between weekdays, weekends, and public holidays in NSW?
3. Can an ensemble regression model predict 30-minute NSW grid demand with a MAPE below 5% on held-out test data?

To answer these questions, we sourced and merged half-hourly AEMO data with weather, rooftop solar, and economic indicators, then ran exploratory analysis to identify key demand drivers. We built and compared Ridge Regression, XGBoost, and CatBoost models on a 60:20:20 chronological split, evaluating each on RMSE, MAE, MAPE, and Adjusted R^2.

Methodology

2 Methodology

2.1 Dataset

2.1.1 Data Acquisition

While the raw AEMO market data provided a high-resolution temporal framework, it lacked the exogenous environmental and economic features crucial for predictive modelling. Consequently, these time-stamps were utilised as the primary keys to synthetically align and merge weather and economic datasets into a unified analytical feature set.

Table 1. Data Collation and Source Rationale

Data Source	Description	Methodology Rationale
AEMO (Market Data)	Half-hourly TOTAL_DEMAND and RRP (Regional Reference Price) for the NSW1 region.	Established the target variable and allowed for the analysis of price-demand elasticity during peak events.
AEMO (Solar Data)	Estimated ROOFTOP_SOLAR_MW generation across NSW.	Critical for deriving GROSS_DEMAND_MW, accounting for "behind-the-meter" generation that masks true load.
Bureau of Meteorology	Half-hourly temperature, humidity, wind speed, and cloud cover data for Sydney.	Included to model the primary driver of demand: the non-linear relationship between ambient temperature and HVAC load.
Economic Indicators	Monthly Cash Rate (RBA), Household Spend (ABS), and Consumer Confidence indices (ANZ-Roy Morgan Australian).	Integrated to capture macro-trends and shifts in baseline energy consumption patterns over the 4-year study period.

2.1.2 Data Cleaning

Data cleaning was limited to linear interpolation of missing weather observations and the alignment of disparate 30-minute and monthly sampling frequencies into a single dataset. Features with high multicollinearity were removed or singled out.

2.1.3 Exploratory Data Analysis (EDA)

Given the conciseness of this report, please refer to Appendix Section A for a comprehensive EDA.

2.1.4 Feature Scaling

To enable the models to better generalise the dataset and remove individuality in entries, the following feature scaling processes were completed:

- **Feature Engineering:** Given the time-based nature of the dataset, the hour, minute, weekend and month were extracted from the `DATE_TIME` feature.
- **Data Transformation:** One-Hot Encoding (OHE) was applied to categorical columns, standard scaling for numerical columns and binary 0/1 for boolean features.

2.2 Technologies Used

The project was executed in a Python 3.11 virtual environment, BeautifulSoup for web scraping, Pandas for data-frames, CatBoost and XGBoost libraries, scikit-learn, and Vegas-Altair for visualisation. Seeding was set to 42 for consistency and reproducibility. This resulted in 70,176 rows across 28 features. Moreover, the 60:20:20 chronological split provided 42105:14035:14036 rows for training, validation and testing respectively.

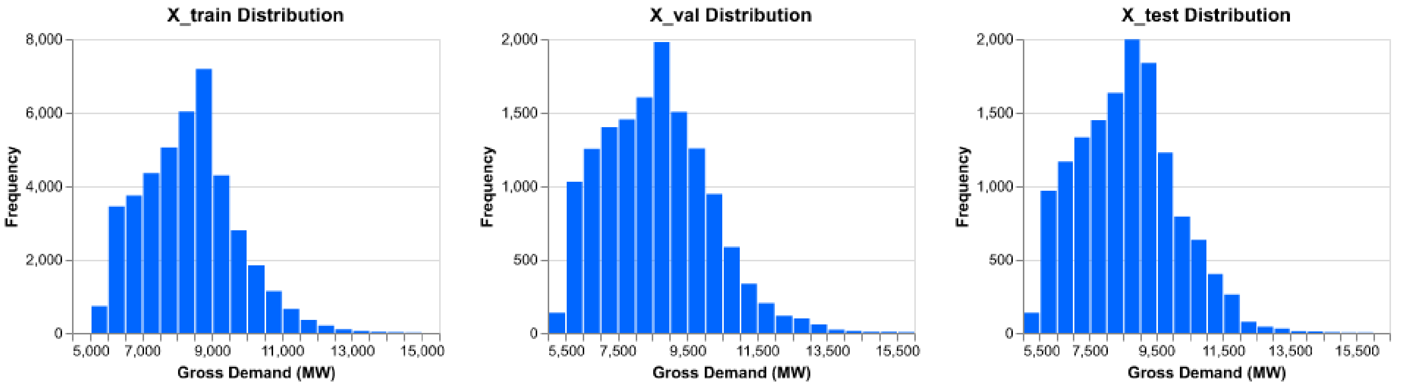

Figure 1. Target feature distribution

2.3 Models

2.3.1 Linear Regression (Ridge)

The linear regressor is a parametric model that fits a linear equation over observational data between a target feature and multiple independent predictors. It served as the baseline noting its computational efficiency and simplicity in capturing non-linear demand patterns. The Ridge variant was chosen to provide a stable baseline by focusing on L2 regularisation to mitigate multicollinearity among highly correlated weather features.

2.3.2 Gradient Boosting Decision Trees

Gradient Boosted Decision Trees (GBDT) are a non-parametric ensemble learning technique that sequentially builds decision trees, where each successive learner is optimised via gradient descent to minimise the residual loss of the previous ensemble.

Table 2. Model Selection and Theoretical Rationale

Model	Description	Methodology Rationale
XGBoost	Scalable GBDT library designed for high computational efficiency.	Selected to address non-linear relationships in the data through sequential tree-building and dual regularisation (L1 and L2) to control model complexity.
CatBoost	Advanced GBDT algorithm that utilises symmetric trees and ordered boosting to reduce prediction shift and handle categorical features.	Chosen for its superior ability to generalise on unseen data and its efficient handling of categorical features (such as 'Season' and 'Holiday') without extensive manual encoding.

2.4 Grid Search

Hyperparameter optimisation is performed via Grid Search with k-fold Cross-Validation. Through exhaustive predefined parameter grids, we identified configurations that minimised generalisation error of each model.

2.5 Metrics

Table 3. Chosen Performance Metrics

Metric	Description and Purpose	Formula
MAE	Average absolute error in MW, operationally interpretable.	$\frac{1}{n}\sum\lvert y_i - \hat{y}_i\rvert$
MAPE	Scale-independent accuracy across demand magnitudes.	$\frac{1}{n}\sum\left\lvert\frac{y_i-\hat{y}_i}{y_i}\right\rvert \times 100$
RMSE	Penalises large outliers in peak demand periods.	$\sqrt{\frac{1}{n}\sum(y_i-\hat{y}_i)^2}$
Adjusted R^2	Explained variance penalised for predictor count.	$1-\frac{(1-R^2)(n-1)}{n-p-1}$

Model performance was evaluated using a hold-out test set to ensure generalisability. The following Table 3 above illustrates the chosen metrics.

Results

3 Results

3.1 Key Findings

CatBoost achieved the best test performance (MAPE 4.47%, Adj. R^2 = 0.872, MAE = 400.25 MW), meeting the sub-5% accuracy target. Surprisingly, Ridge Regression explained only 38.9% of demand variation despite regularisation, confirming NSW electricity demand is fundamentally non-linear.

3.2 In-depth Results

3.2.1 Can an ensemble regression model predict 30-minute NSW grid demand with MAPE below 5%?

Three models were compared on held-out test data (Table 1). Ridge Regression failed to capture non-linear patterns (MAPE 10.31%, R^2 = 0.389). XGBoost improved but showed overfitting (MAPE 5.83%, R^2 = 0.836). CatBoost generalised best across all metrics, meeting the sub-5% threshold. Regression assumptions were verified via QQ-plots (shown in Figure 2).

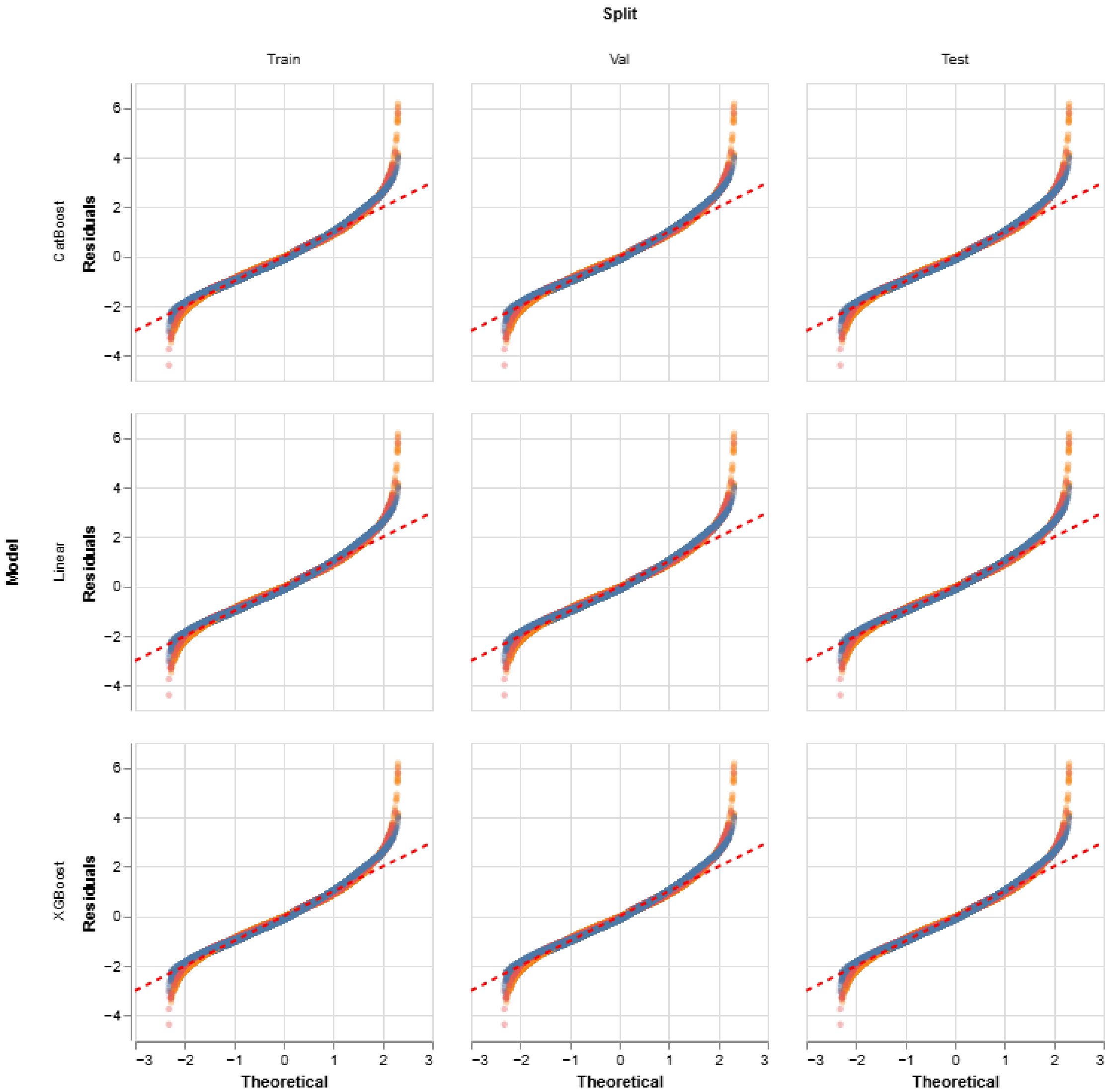

Figure 2. QQ Plots

Conclusion

4 Conclusion

4.1 Conclusion

4.2 Discussion

4.3 Project Limitations and Caveats

4.4 Stakeholder Analysis and Project Outcomes

References

References

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Appendices

Appendices

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A Exploratory Data Analysis

The objective of this exploratory phase was to move beyond the raw complexity to identify the primary drivers of Gross Energy Demand. In light of the data science project cycle, this EDA served two critical functions: **Feature Selection** and **Operational Insights**.

A.1 General Exploration

Observing multicollinearity such as through a correlation heatmap is critical because it identifies highly redundant predictors that can destabilise the model, inflate the variance of coefficient estimates, and make it impossible to determine the individual mathematical contribution of each variable to the target output.

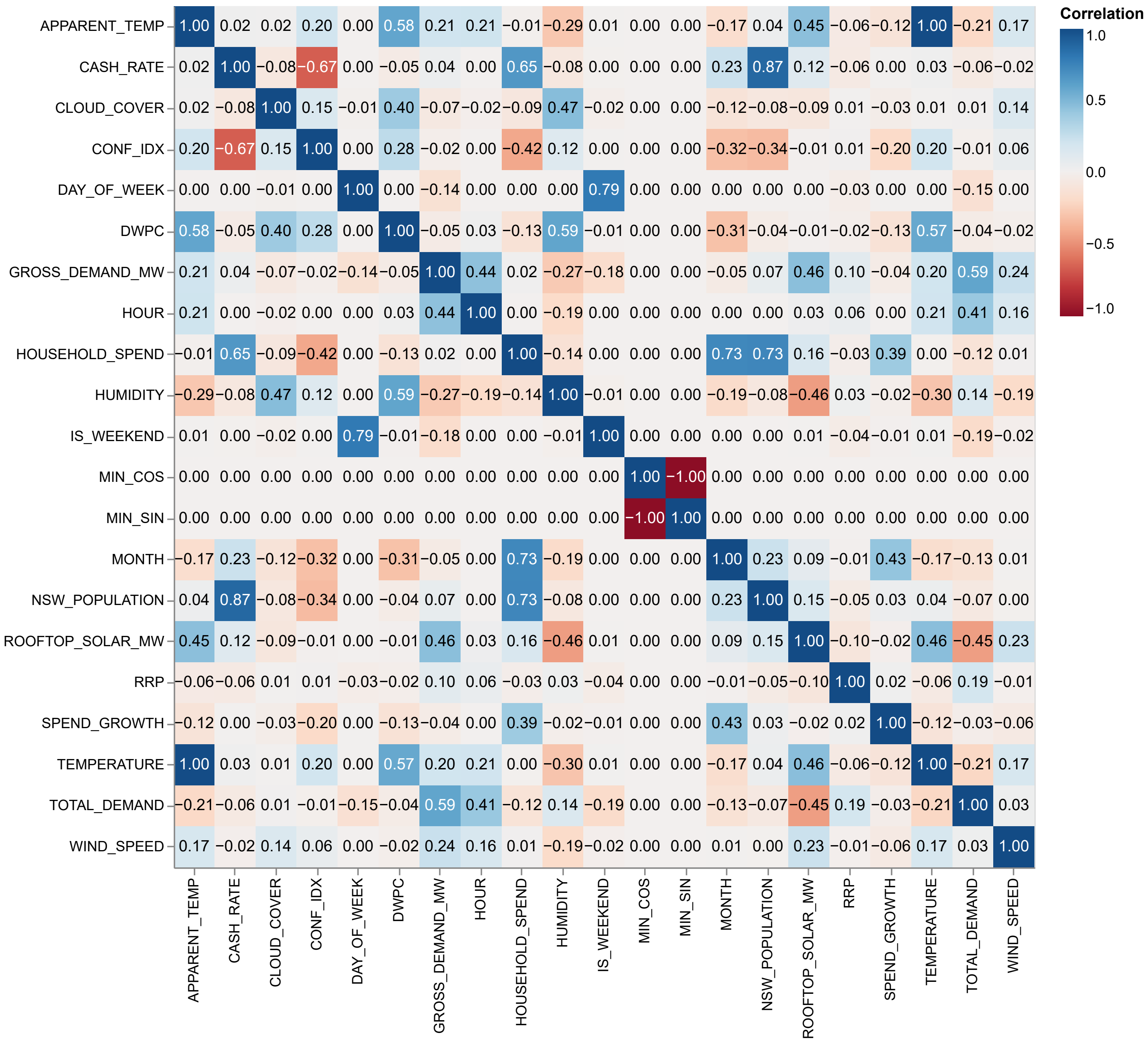

Figure 3. Multicollinearity Matrix on Raw Data

A.2 Does temperature have a non-linear relationship with NSW grid demand across 2°C bins?

A clear U-shaped relationship was observed between temperature and mean grid demand (Figure 4). Demand was lowest near 21°C and rose sharply at both cold and hot extremes, directly ruling out linear modelling.

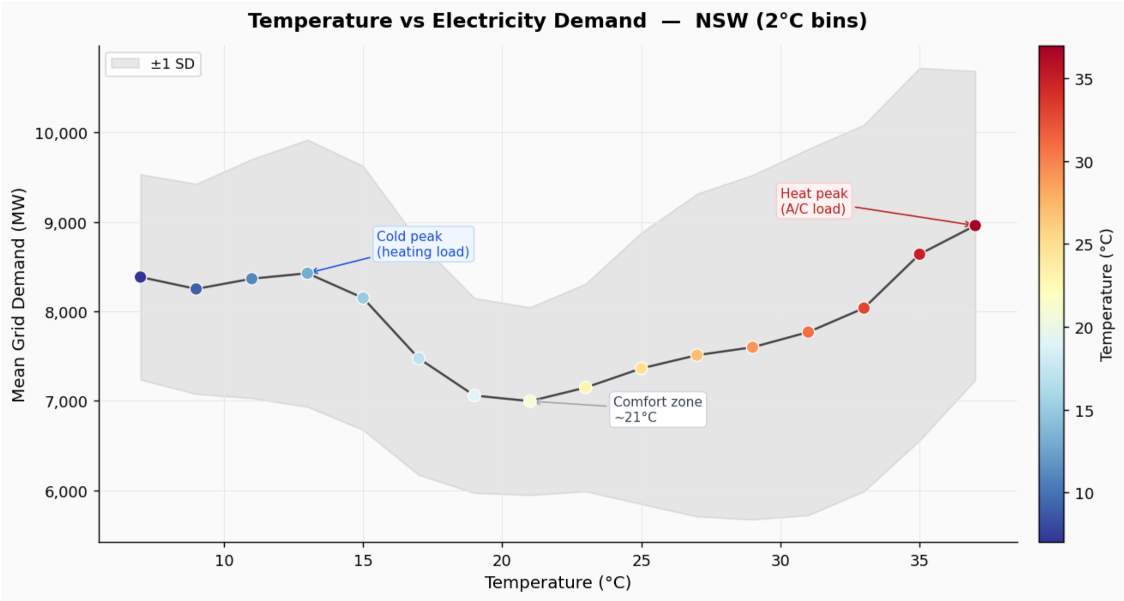

Figure 4. Mean NSW grid electricity demand (MW) across 2°C temperature bins (5°C-43°C). A U-shaped pattern is evident, with lowest demand near the comfort zone (21°C, 7,000 MW), rising at cold extremes due to heating load and hot extremes due to air-conditioning. Shaded band represents ±1 SD.

A.3 Do mean hourly demand profiles differ between weekdays, weekends, and public holidays?

Weekday demand showed a morning commercial peak at 8 AM (8,200 MW) and an evening residential peak at 7 PM (9,500 MW). Weekend and holiday profiles were 1,000-1,500 MW lower with no morning peak (Figure 2). An unexpected finding was a pronounced midday dip across all day types driven by rooftop solar, forming the classic "duck curve" pattern (Figure 5).

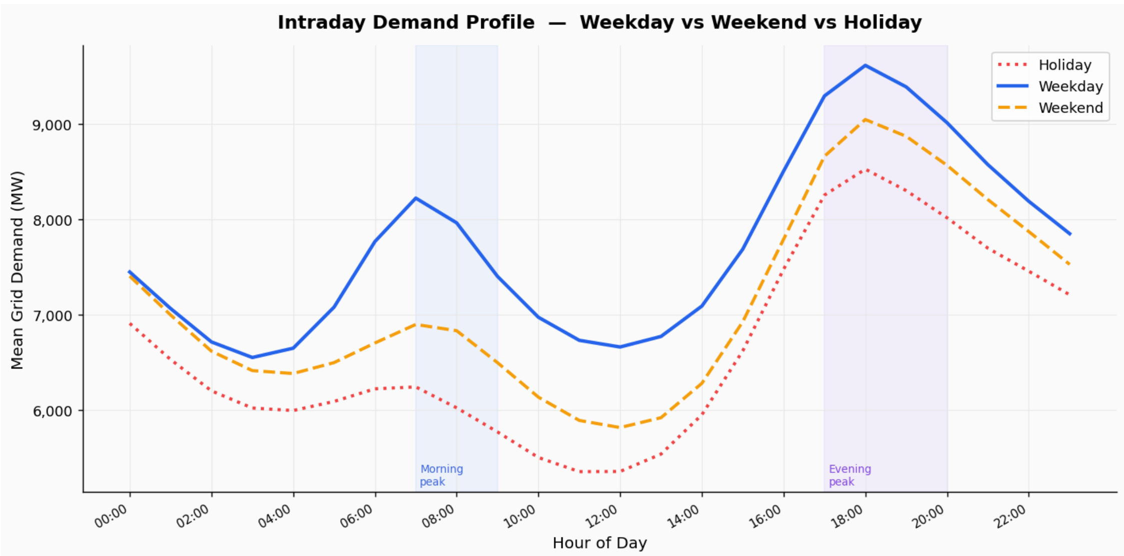

Figure 5. Mean NSW grid electricity demand (MW) by hour of day for weekdays (solid blue), weekends (dashed orange), and public holidays (dotted red). Weekdays show a morning peak at 8 AM (8,200 MW) and an evening peak at 7 PM (9,500 MW). Weekend and holiday profiles are 1,000–1,500 MW lower with no commercial morning peak.

(Figure 6)

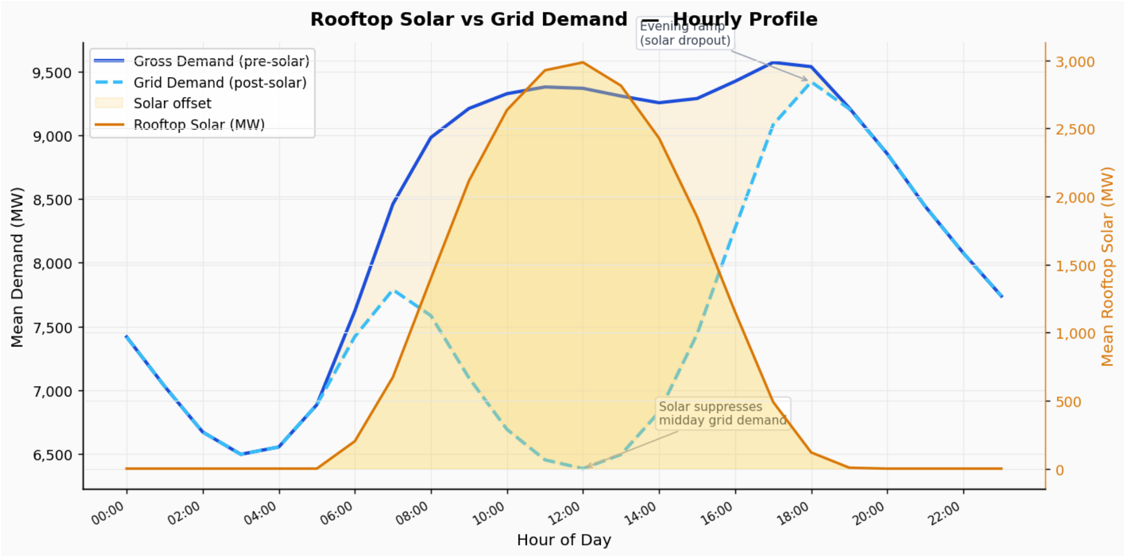

Figure 6. Mean hourly rooftop solar generation (MW, right axis, orange) overlaid on gross pre-solar demand (solid blue) and net post-solar grid demand (dashed teal). Solar peaks near midday (3,000 MW), suppressing grid demand and creating the sharp evening ramp as generation drops — the "duck curve" pattern.

B Modelling Code